RareDiseaseAI Platform

System Architecture, Pipeline Workflow, & Features Overview

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1. EXECUTIVE SUMMARY

The **RareDiseaseAI Platform** is an international-grade, secure, multimodal diagnostic clinical decision support system. It integrates high-performance artificial intelligence algorithms with professional medical workflows to ingest, process, and analyze heterogeneous patient data—including **MRI/CT Scans, Genomic Variants, Clinical Notes, and Laboratory Panels**.

By merging these disparate inputs, the platform makes a joint ensemble prediction on potential rare disease conditions, furnishing clinicians with multiple levels of Explainable AI (XAI): modality contribution weights, feature relationships, and interactive regions of interest (ROI) overlays directly on scans.

2. CORE PIPELINE WORKFLOW (STEPS 1 - 9)

The diagnostic engine runs inside a background worker queue divided into 9 deterministic, fault-tolerant execution stages:

- **Step 1: Patient Intake Form:** Gathers patient demographics (Patient ID, Name, Age, Biological Sex, suspected condition) and clinician details.
- **Step 2: Data Quality & Dependency Validation:** Enforces data integrity checks (e.g. imaging file is required, and at least 2 other optional files must be uploaded).
- **Step 3: Parallel Feature Extraction:** Processes multiple inputs concurrently using Python's ThreadPoolExecutor. Extracts DICOM/NIfTI slices, evaluates genetic graphs, executes BioBERT-derived transformer embedding on clinical notes, and computes normalized z-scores on lab inputs.
- **Step 4: Ensemble ML Inference:** Issues a request to the deep-learning classifier model to yield a disease prediction, confidence level, attention weight arrays, and textual reasoning.
- **Step 5: Explanation & Scan Annotation:** Generates the modality influence chart and constructs disease-specific highlight (ROI) masks (like Caudate Nucleus regions for Huntington's or Lung Infiltrates for CF) onto the imaging scan.
- **Step 6: Diagnostic PDF Compilation:** Generates a styled, publication-ready PDF report detailing the diagnostic outputs, next steps, contribution graph, and annotated scan.
- **Step 7: Database Registration:** Persists all diagnostic parameters, metadata, and artifact file paths to a PostgreSQL database.
- **Step 8: Email Notification:** Prepares a formal diagnostic notification email containing links to download the report, along with the PDF and chart attachments, saved as a backup `.eml` file.
- **Step 9: Hospital Dashboard Sync:** Performs an API callback to sync local hospital dashboards and client-side tables with the completed diagnostic results.

3. COMPLETED PREMIUM FEATURES

A. Doctor & Administrator Portals

- **Doctor Authentication (/login.html):** Features a sleek dual-pane layout, dark glassmorphism, floating label inputs, and demo credential assistance.
- **SuperAdmin Dashboard (/admin.html):** Offers sidebar tabbed control, real-time analytics indicators, registered doctor tables with status toggles and delete capabilities, doctor register form, and a system activity feed.

B. Advanced Diagnostic Views

- **Interactive Scan Inspector:** Displays custom anatomical overlays highlighting diagnostic pathology. Equipped with real-time brightness/contrast range sliders and a fast reset button.
- **XAI Knowledge Network:** Powered by Vis.js, this displays interactive node models showing how different modalities contribute to the final AI diagnostic judgment.
- **Global Registry Map:** Renders simplified disease registry prevalence hotspots globally.
- **Telehealth Collaboration Hub:** A simulated workspace to request a peer review board, activating automated clinical discussion between specialists (Dr. Schmidt in Berlin and Dr. Tanaka in Tokyo).
- **History Dashboard Recall:** Added a 'View Dashboard' action in the history table to reload previous analyses instantly into the active dashboard.

4. SYSTEM FILE INVENTORY & UPGRADES

File Path	Implemented Upgrades & Changes
app.py	Added auth session management, admin register endpoints, system stats metrics, and static serves.
pipeline_annotated.py	Custom overlay modules. Overrides Step 5 and 6 to produce scan ROI images and attach them to the ReportLab layout.
pdf_report.py	Aesthetic ReportLab stylesheet configurations. Embeds annotation panels and lists modality findings.
static/login.html	Doctor/Admin login workspace styled with responsive glassmorphism.
static/admin.html	SuperAdmin registry and statistics monitor.
static/index.html	Intake form and diagnostics layout redesigned into a tabbed Single Page App (SPA).
static/styles.css	Unified dark slate, royal indigo, and emerald teal color system with micro-animations.
static/app.js	Client controller code for pipeline status polling, rendering Vis.js graph networks, and collaboration chat logs.
static/doctors.json	Clinician registry seed file.

5. OPERATING INSTRUCTIONS

To access the local deployment instances, use the following browser endpoints:

- **Main Application:** http://localhost:5000/ (Log in with *sarah.chen@hospital.com* / *Doctor@123*)
- **SuperAdmin Management:** http://localhost:5000/admin.html (Log in with *admin@raredisease.ai* / *Admin@123*)
- **Integration Test Suite:** Run *python test_pipeline.py* from the project workspace to assert database, files, and annotations.